Aman Panday

Contact No: 6284386155 | Amritsar, India | Email: amansharmaasr00@gmail.com

LinkedIn:https://www.linkedin.com/in/amansharrma00/ | GitHub:https://github.com/AmanSharrma00

CAREER OBJECTIVE

Optimistic, enthusiastic, and growth-driven Engineer who **ships and owns production systems**, not just prototypes. **Built and deployed** a MERN e-commerce platform end-to-end, engineered a Java/JDBC/MySQL system with **ACID-compliant transactions** and RBAC, and architected an **IoT + Computer Vision pipeline** that **cut manual monitoring effort by ~70% and irrigation water usage by ~25%**. Comfortable across the application, data, and AI layers, with hands-on experience in secure authentication, data integrity, **LLM prompt engineering, and agentic workflows**.

TRAINING & EPERIENCE

Cyber-Physical Systems & IoT, — *In-House Program, Chandigarh University* 05/2025 – 06/2025 | India

- **Completed** a 6-week in-house program on cyber-physical systems, **building** hands-on with sensor-actuator integration and embedded IoT design.

EDUCATION

Chandigarh University, *Bachelors of Engineering - CSE | CGPA: 6.72/10* 06/2023 – Present | Mohali, India

Khalsa College, *Diploma in Computer Application | Score: 8.3/10* 08/2022 – 08/2023 | Amritsar, India

CORE SKILLS

- **Language & Core Foundation ::** Python (core syntax, scripting , Pandas, Numpy, Flask) | (Java (OOP, DSA —Array, String)
- **AI & ML::** TensorFlow, Neural Networks (weights, bias, forward/backpropagation, loss functions) | CV model benchmarking (CNN, YOLOv8, Mask R-CNN) | Generative AI & LLM Prompt Engineering (structured output, role prompting) | Agentic/Tool-Use
- **Databases & Data Integrity::** MongoDB Atlas | MySQL(joins, aggregation, schema design, indexing) | JDBC | ACID transactions
- **Cloud & DevOps ::** Git & GitHub | Firebase Hosting | Render | CI&CD | AWS
- **Full-Stack::** React.js | Node.js | Express.js | JavaScript (ES6+) | HTML5 | CSS3 | React Query | JWT Authentication | REST APIs
- **CS FUNDAMENTALS: Operating Systems ::** Process/Thread management, Scheduling, Memory Management, Deadlocks, Concurrency | **Computer Networks ::** OSI/TCP-IP Model, HTTP/HTTPS, DNS, TCP vs UDP, REST API communication

INTRA-PERSONAL SKILLS

Problem Solving: Solved real-worldchallenges through structured problem-solving, sharpened by consistent DSA practice.

Self Learning: Self-taught the MERN stack and IoT development end-to-end, demonstrating fast independent learning.

Time Management & Leadership: Led the college cricket team as captain while balancing academics and project deadlines.

Team Orientation: Led the college cricket team as captain, coordinating decisions under match pressure

PROJECTS

AI-Driven Data-Entry Automation Bot, *Python, LangChain, Selenium, Flask* 12/2025 – Present

- Building an extraction-to-structuring pipeline using pdfplumber for PDF invoice parsing and a **LangChain + Groq (Llama 3.1) chain with a Pydantic-enforced schema** to convert raw text into structured JSON (vendor, amount, date, category).
- Implementing a **Selenium-driven form-filling module** that combines rule-based web automation with LLM-based field classification .
- Architecting for deployment on **Render (Flask + Gunicorn)** with **MongoDB Atlas** for document-based storage of structured invoice data; scoped the browser-automation piece to run locally due to headless-Chrome limitations on free-tier hosting.

SAHAYAK – Full Stack E-commerce Platform, *MERN STACK (DEPLOYED)* 11/2025 – 06/2026

- **Owned the product lifecycle end-to-end** — requirement analysis, data schema design, REST API build, and production deployment on Render/Firebase — **cutting rework by validating requirements before implementation.**
- Engineered role-based access control and multi-layer validation across buyer/seller workflows; **diagnosed and resolved post-launch operational issues** to sustain uptime under real user load.

SMART AGRO-CARE, *IoT + AI Research Project (ESP32 / CNN, YOLOv8, Mask R-CNN)* 2023 – 2024

- **Benchmarked 3 CV architectures** (CNN, YOLOv8, Mask R-CNN) head-to-head on accuracy/latency trade-offs and defended the evidence-based model choice to a faculty panel — **placed Top 10 of 200+ teams** at CU Project Expo across all 3 evaluation rounds.
- Architected a real-time sensor-to-decision pipeline that **cut manual monitoring effort by ~70%, reduced irrigation water usage by ~25%,** and **lifted crop yield 20–25%.**

CERTIFICATION

- Build a computer vision app with Azure Cognitive Services
- The Arduino Platform and C Programming - Coursera
- IoT Devices - Coursera
- Data Analysis_Assessment

AWARD & ACHIEVEMENT

CU PROJECT EXPO TOP 10 out of 200+ teams, Levels 1, 2 & 3, *Chandigarh University* 2024

ACP Rank 1st University-wide, *Chandigarh University* 2023